

ESP Introduction Training

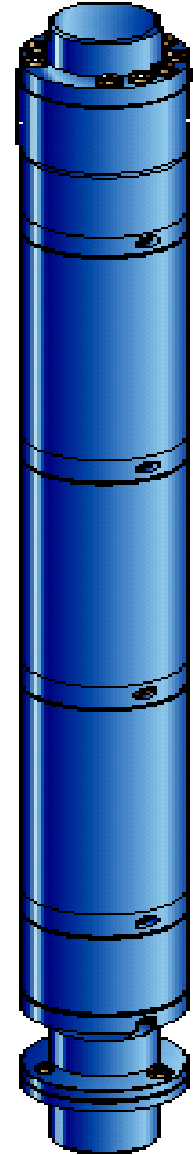
Day 2

Protectors

Protector Functions

The Protector has four primary functions:

- Prevents entry of well fluid into the motor
- Provides pressure equalization to the motor
- Houses the thrust bearing to carry the downthrust developed by the pump
- Couples the torque developed in the motor to the pump via the protector



Protector Chambers: Labyrinth

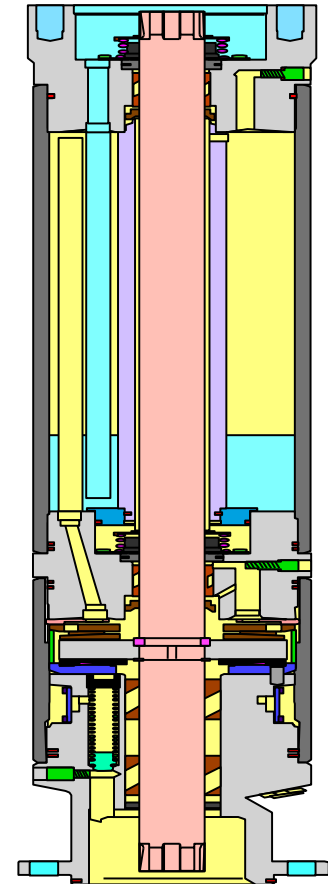
Labyrinth chamber relies on difference in SG between motor oil and well fluids

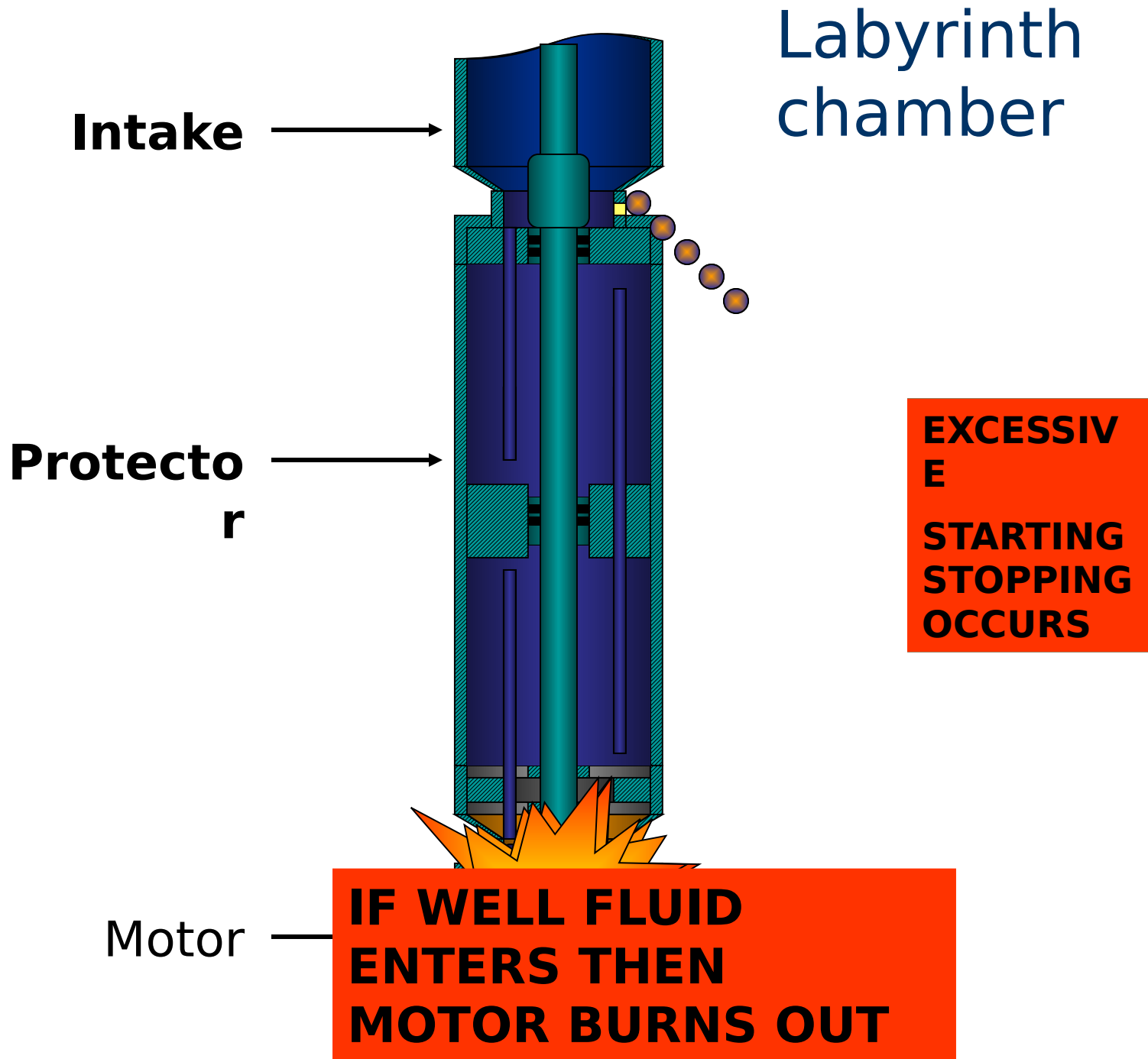
Advantages

- Good in vertical wells
- Easily serviceable and re-usable
- Resistant to temperature and chemicals
- Inexpensive

Disadvantages

- Not applicable to highly deviated wells
- Not applicable if oil SG < 0.85SG (35 °API)





Protector Chambers: Bag

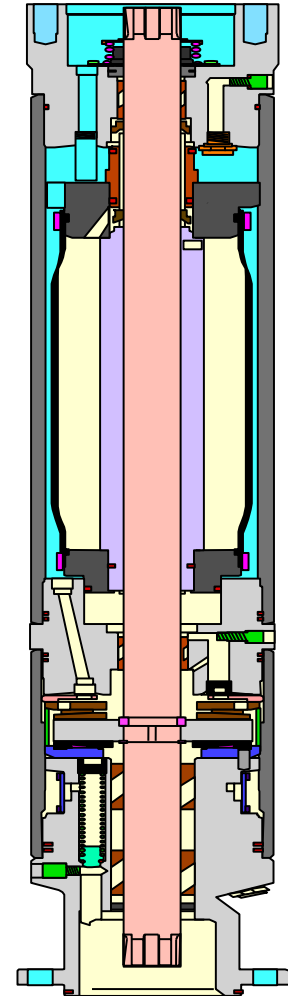
A bag or positive seal provides a physical barrier that separates the clean oil from the contaminating well fluids.

Advantages

- Positive separation of well fluids from motor oil
- Operates in vertical and deviated wells
- Operates regardless of well fluids SG

Disadvantages

- Rubber bags may be affected by temperature, solids and chemicals
- Re-usage is limited



Bag chamber

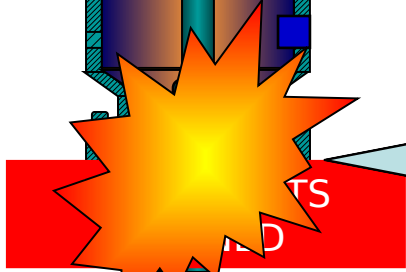
Bags
contracts
due to
getting
less oil
inside

3-5 PSI

STOP

Torn or leak
(Could be seal failure)

Temperature decrease



Chamber Connection

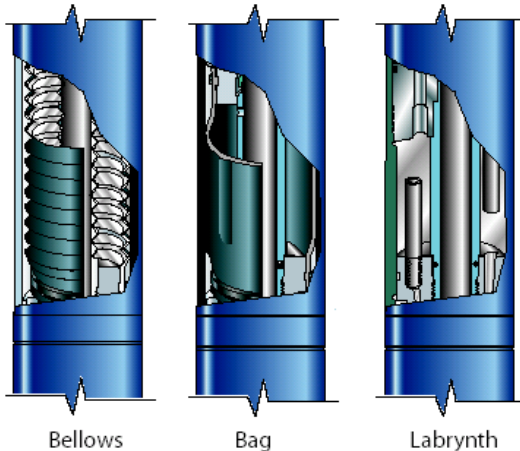
Parallel

- Results in a larger volume capacity, which is necessary for high House Power applications
- Only possible with bags or bellows

Series

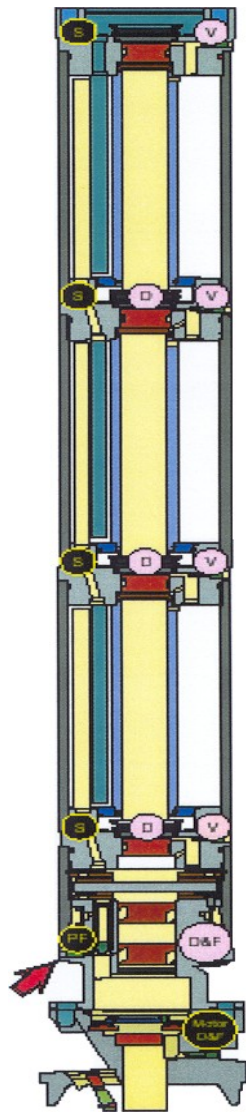
- Results in redundant seals
- Increases the reliability of protector

Protectors – Naming Convention

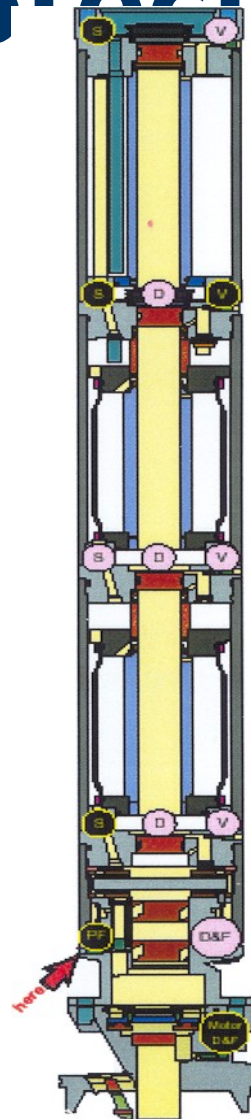


- Modular are a combination of several chamber types
 - “L” type (**L**abyrinth)
 - “B” type (Elastomer **B**ag)
 - “M” type (**M**etal Bellows)
- Which are combined in ...
 - “S” (series) – adds redundancy
 - “P” (parallel) – adds capacity
- Thrust bearings
 - Upthrust/Downthrust (KTB/HL)

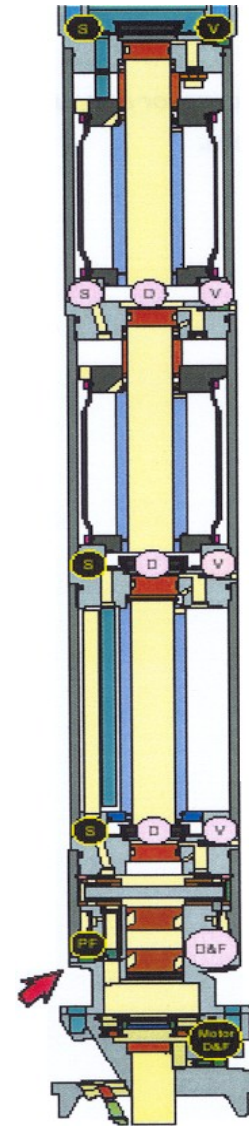
Modular Protectors



LSLSL

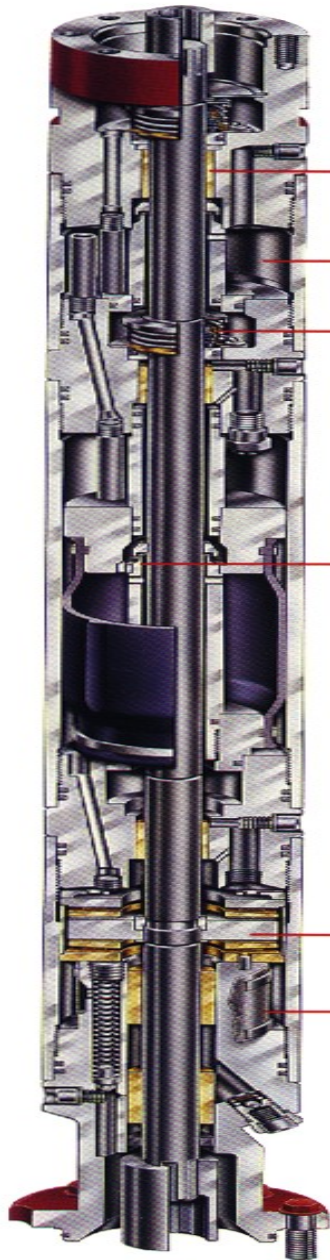


LSBPB



BPBSL

Protector Components



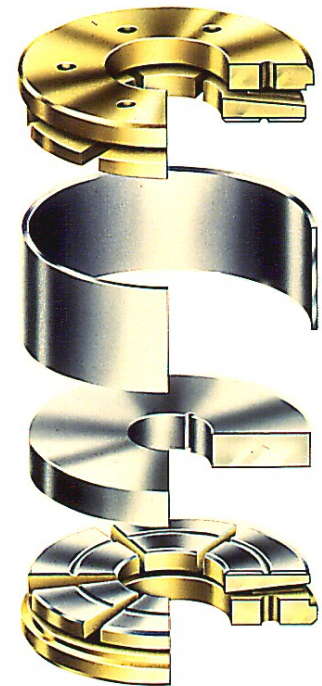
Bearing

Shaft Seal

Bag Chamber

Thrust Bearing

Filter

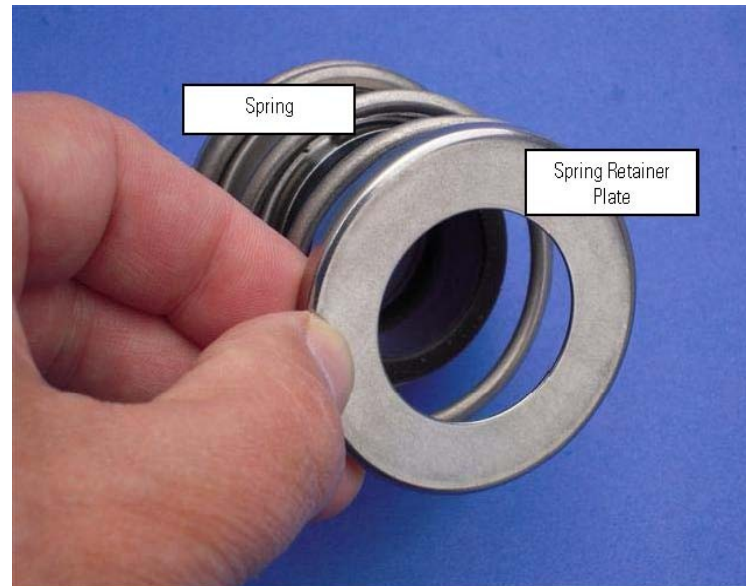


Mechanical Seals

Prevent leak of fluids in between chambers through the space between shaft and protector body (seal body)

Seals main parts

- Silicon carbide sealing washer and seal seat
- Monel spring
- HSN/Aflas bellows



Protector Metallurgy

When choosing Protector metallurgy, use the same guidelines as for pumps.

Housing

- Carbon Steel (1026 CS)
- Redalloy (9Cr, 1Mo)
 - Upgrades: 316 SS, 22Cr Duplex

Shaft

- Monel K500
- HS Monel
- Inconel 625
- Inconel 718

Head and Base

- Carbon Steel (8620 CS)
- Stainless Steel (416 SS)

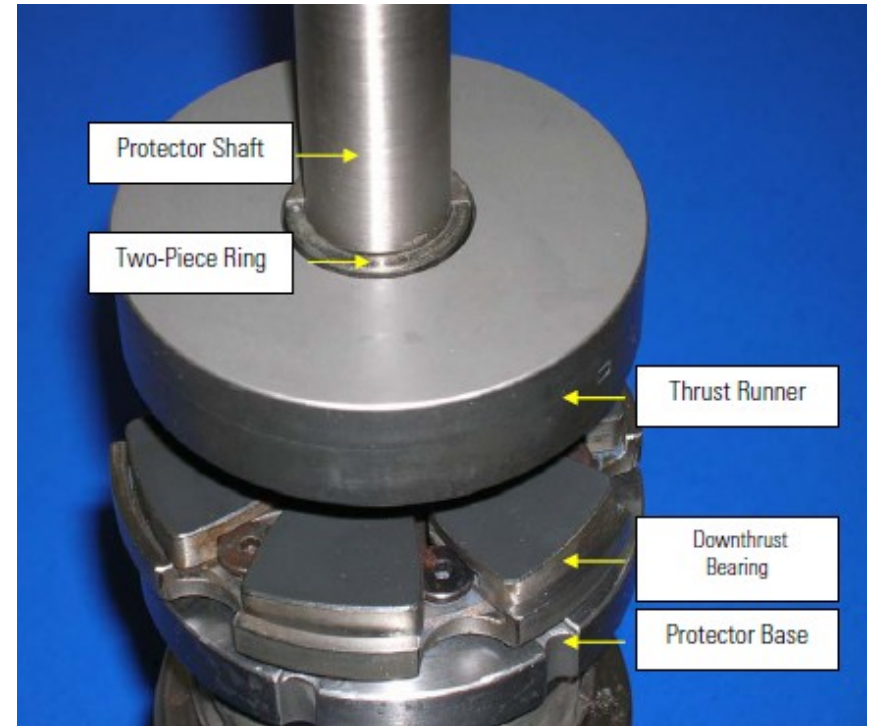
Elastomers

- HSN
- Aflas – Higher temperature and chemicals resistance

Environment:	Elastomer families - Qualified compounds only*						
	Neoprene® fiber reinforced	Nitrile	HNBR	Viton®	Aflas®	Chemraz®	Kalrez®
Maximum recommended temperature	275°F	275°F	325°F	400°F	450°F	550°F	550°F
Minimum recommended temperature	75°F	minus 40°F	minus 40°F	40°F	75°F	40°F	75°F
Crude oil	OK	OK	OK	OK	OK	OK	OK
Natural gas w/ condensate	OK	OK	OK	OK	OK	OK	OK
Formation or injected water	OK	Up to 275°F	Up to 325°F	Up to 300°F	OK	OK	OK
H ₂ S	NO	Up to 10 ppm	Up to 100 ppm	Up to 375°F	OK	OK	OK
CO ₂ Gas	OK	OK	OK	OK	OK	OK	OK
Water based mud	Up to 275°F	Up to 275°F	Up to 325°F	Up to 300°F	OK	OK	OK
Oil based mud	OK	OK	OK	OK	OK	OK	OK
Ester containing mud	NO	NO	NO	OK	OK	OK	OK
Diesel	OK	OK	OK	OK	OK	OK	OK

Thrust Bearing

- Thrust generated by the pump determines thrust bearing capability.
- Down thrust at zero flow is to be considered as it is the “worst case scenario”



Thrust Bearing

When selecting the thrust bearing, you must consider:

- Temperature of operation
- The oil used in the protector
- The load rating

Thrust Bearing

Abbreviation	Full Name	Properties
STB	REDA Babbitt	• Bi-Directional • Low temperature
NTB	REDA Bronze	• Bi-Directional • Higher temperature
KTB	KMC Bronze	• Bi-Directional • Higher load • Higher temperature
HL	High Load Tilt Pad	• Uni-Directional • Highest load • Higher temperature



STB

NTB

KTB

HL

Protector Oil

Installation

- Refer to ALFORM-D for correct procedure
- Minimize air bubbles
- Same oil as used in motor

Purposes

- Cool the motor and protector
- Lubrication
- Increases efficiency/reduces power loss
- Provides pressure continuity/equilibrium

Protector Oil

Motor Internal Temperature (F / C)	Oil Name	Comments
< 270 / 130	Reda # 3	Cold well applications
< 290 / 145	Reda # 2	Mineral Oil, recommended for production of drinking water
< 360 / 180	Reda # 5	Default Motor oil in Dominator and Maximus Series
< 400 / 204	Reda # 6	Viscous oil that requires motor internal temperature to be in between 300 and 400 F (150 to 204 C)
>400 / 204	Reda # 7	SAGD, Hotline Motors